Prospective source model of the 2026 M_w 7.3 Mexico (Chiapas) earthquake

A blind test of our MTUQ+WISP pipeline, produced before the USGS finite fault

Automated pipeline (17.Venezuela_2026)

Frozen 2026-07-18, hours after origin — for later comparison with the USGS solution

Purpose

The USGS had not yet posted a finite-fault model for `us7000t1bu` when this was produced. This is a prospective (blind) test: we run our teleseismic MTUQ moment-tensor + WISP finite-fault pipeline on open GSN data and freeze the prediction here so it can be compared with the USGS solution once posted. Everything below uses only teleseismic body + surface waves and the USGS W-phase moment tensor as the seed; no InSAR, regional, or geodetic data (which USGS will likely add, and which — per our Al Haouz study — tends to yield a smaller, more compact source).

1 Event and USGS preliminary benchmark (frozen)

Origin	2026-07-17T14:48:39.77 UTC, 14.604°N, 92.952°W, depth 18.6 km
Magnitude	M_w 7.3 (W-phase); $M_0 = 1.1 \times 10^{20}$ N m; centroid depth 15.5 km
Tectonics	Middle America subduction zone (Cocos plate), shallow thrust
W-phase NP1 (steep)	strike 122.75, dip 63.86, rake 97.74
W-phase NP2 (shallow)	strike 285.6, dip 27.19, rake 74.65 (likely megathrust interface)

This is analogous in size to the 2024/2026 Atacama and 2026 Venezuela events we have modelled, but it is a shallow subduction thrust, so we apply the shallow-source lessons from our Al Haouz (Morocco) study.

2 Data

23 GSN broadband stations (networks IU, II, G, GE, …) at 31–86°, fetched from IRIS. WISP uses 15 with usable P; MTUQ uses 11 three-component stations. The azimuthal coverage has a ∼87° gap over the Pacific (few island stations) — adequate but not ideal.

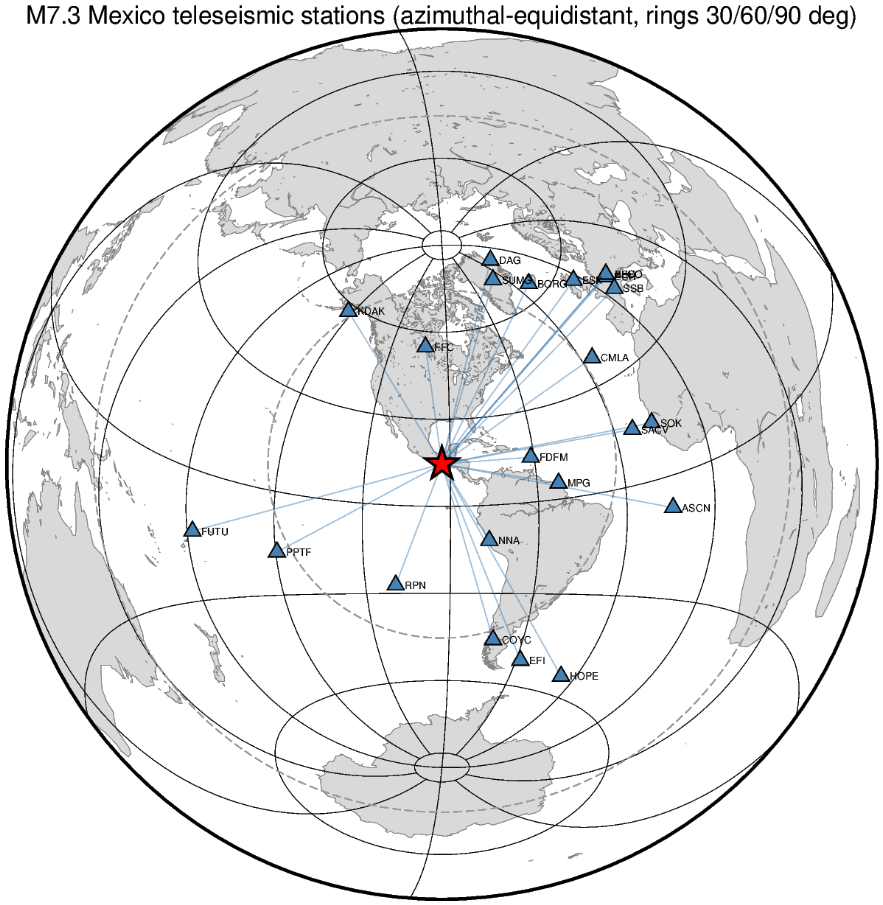

Figure 1: Teleseismic station distribution (azimuthal-equidistant, rings at 30/60/90°).

3 Moment tensor (MTUQ) — the shallow-thrust trap, a third time

For this shallow source the same pattern we documented on Al Haouz reappears, now at M_w 7.3:

MTUQ solution	strike/dip/rake	M_w	verdict
DC, body-only (adopted)	284/30/79	7.30	≈ W-phase NP2 (shallow interface)
DC, surface waves	121/85/88	7.45	dip 85° (too steep — shallow indeterminacy)
full moment tensor	126/81/81	7.60	destabilised ($w = 0.90$, huge non-DC)
W-phase (USGS)	122.75/63.86/98 / 285.6/27.19/75	7.3	reference

The body-only DC recovers the W-phase mechanism and moment (284/30 matches the shallow NP2; M_w 7.30 matches M_{ww}). The surface-wave DC pushes the dip to 85° (the Kanamori–Given shallow dip-slip indeterminacy), and the full moment tensor is destabilised into a large spurious non-double-couple with M_w 7.60. This is the third independent confirmation of our rule: for a shallow source, invert body waves for the mechanism. We adopt the body-only DC.

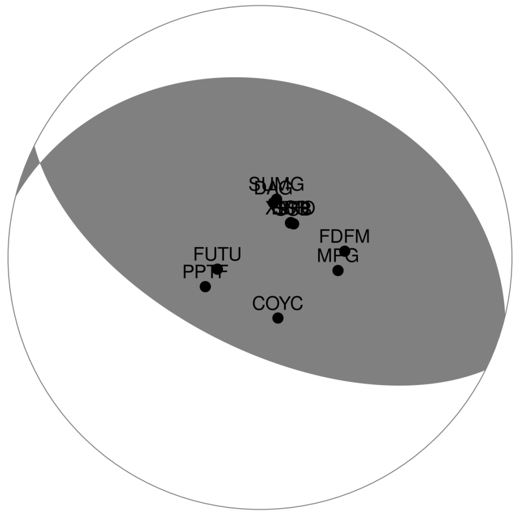

Figure 2: Adopted body-only DC moment tensor (284/30/79) with the station distribution — a shallow subduction thrust, consistent with the W-phase.

4 Finite fault (WISP), both nodal planes

WISP `auto_model` on both W-phase nodal planes, seeded from the W-phase moment (the key lesson from our Al Haouz ablation: seed from the authoritative M_{ww}). Teleseismic body + surface waves.

plane	strike/dip	M_w	misfit	$L \times W$ / peak slip	slip depth / T_{5-95}
NP1 steep	123/64	7.29	0.1120	82×41 km / 2.83 m	1–34 km / 19.9 s
NP2 shallow (interface)	286/27	7.29	0.1114	82×68 km / 2.29 m	3–30 km / 20.8 s

Table 1: Both planes fit almost identically (0.1120 vs 0.1114) — teleseismic waveforms are a weak plane discriminant, as we found for Al Haouz and Chile. For a subduction thrust the shallow-dipping NP2 is the megathrust interface and fits marginally better; we prefer NP2 but show both. Both give M_w 7.29 (matching M_{ww}), a compact updip asperity, and ∼20 s duration.

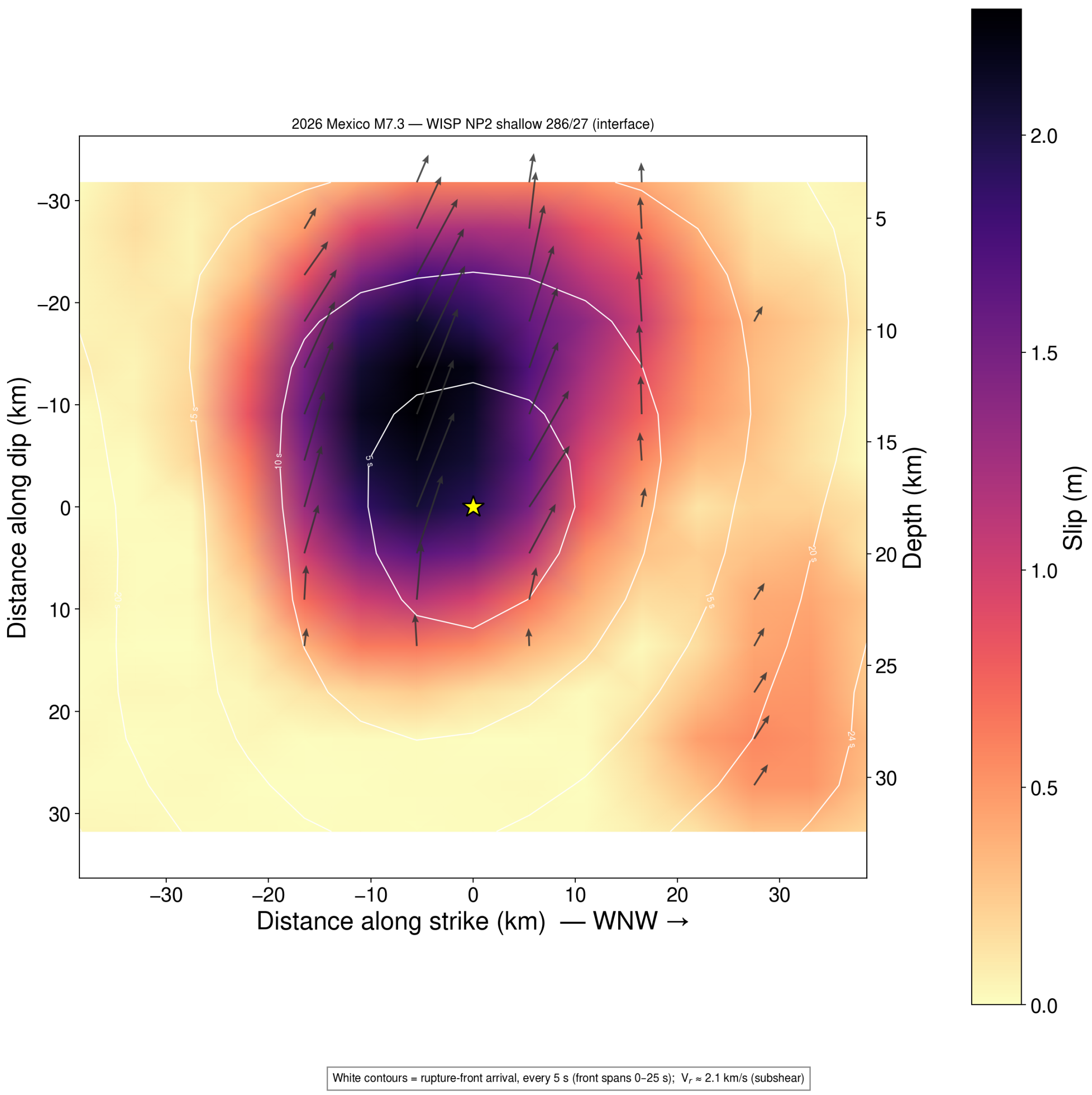

Figure 3: Preferred (NP2, shallow interface) on-fault slip at true 1:1 scale; star = hypocentre, white contours = rupture front every 5 s, arrows = thrust slip. Compact updip asperity, peak 2.3 m, slip ~5–30 km depth.

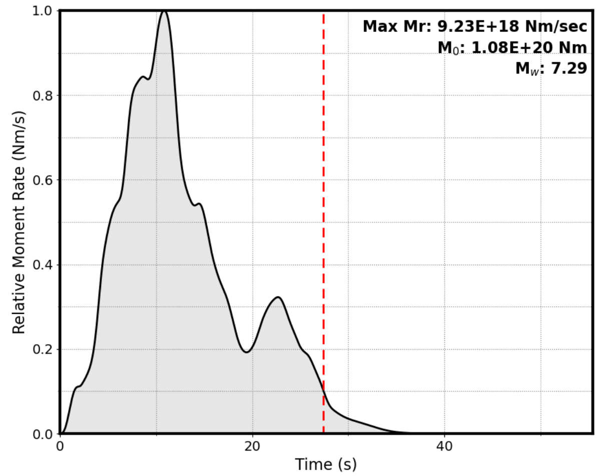

Figure 4: Moment-rate function (NP2): peak at ∼11 s, total duration ∼20 s.

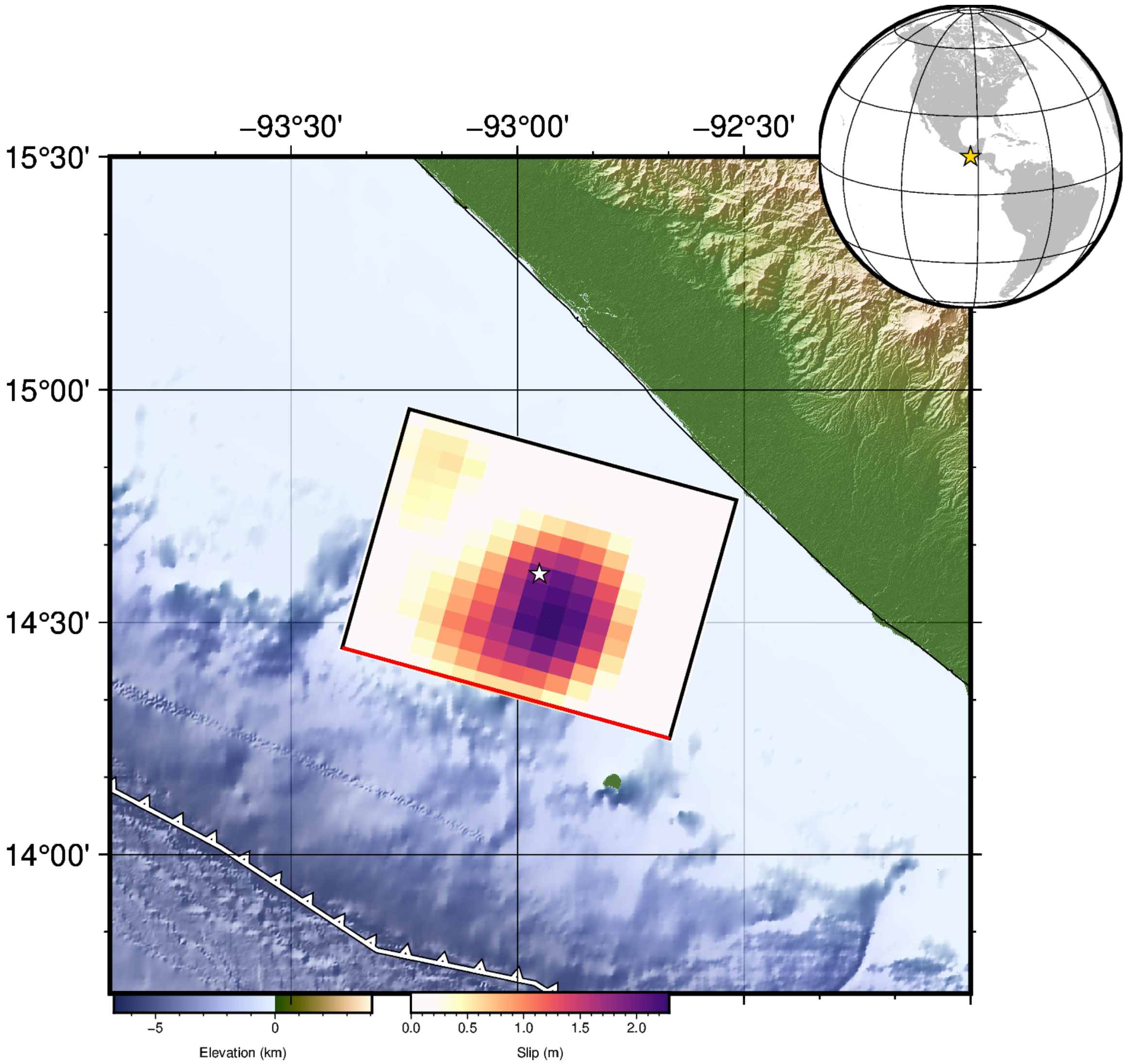

Figure 5: Map view of the NP2 slip projected to the surface.

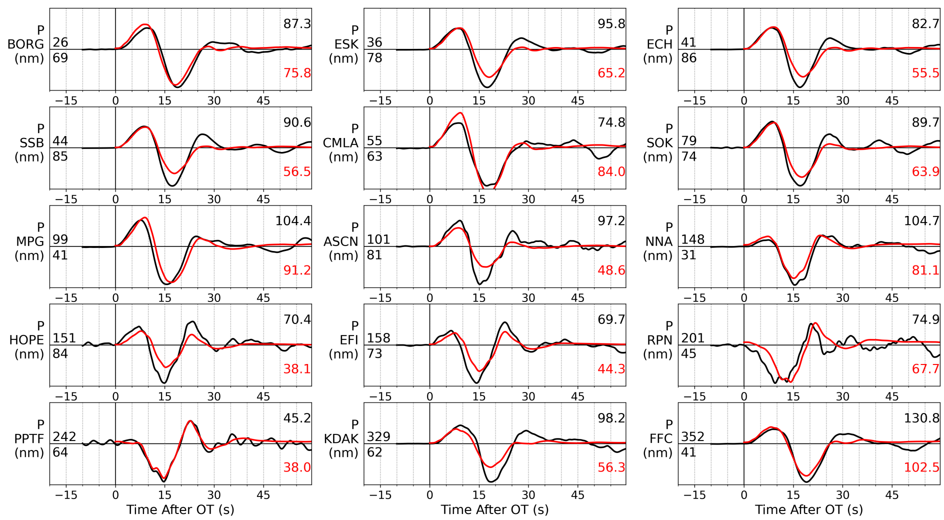

Figure 6: WISP teleseismic P-wave fits (NP2). Black observed, red synthetic.

5 Prospective prediction (compare against the USGS FFM when posted)

- Mechanism: shallow subduction thrust; preferred plane ≈285/27/75 (megathrust interface), or steep conjugate 123/64.
- M_w: 7.29–7.30 ($M_0 \approx 1.1 \times 10^{20}$ N m), matching M_{ww}.
- Peak slip: ∼2.3–2.8 m; slip depth ∼5–30 km; single compact updip asperity near/above the hypocentre.
- Duration: ∼20 s, moment-rate peak at ∼10 s.

Honest caveats. (1) Teleseismic-only: no InSAR/regional yet. Per our Al Haouz study, adding geodesy usually shrinks the source and raises peak slip; our 82-km-long fault may be over-sized for M_w 7.3. (2) Location is the catalogue hypocentre (not relocated to a geodetic centroid). (3) The ∼87° Pacific azimuthal gap limits resolution on the down-dip/along-strike updip direction. These are exactly the axes on which we expect the USGS seismogeodetic model to improve; the mechanism, moment, and first-order slip depth should be robust.